

Question 1: Let $\Sigma = \{a, b\}$. Consider the following language.

$$L1 = \{w \mid w = y\#z, y \in a^{n+1}b^{2n} \text{ and } z \in b^{n+1}a^{n+2}, n \geq 0\}$$

$$L2 = \{w \mid w \text{ a string which is an odd length palindrome}\}$$

$$L3 = \{w_1\#w_2 \mid w_1 \text{ contains odd number of } b\text{'s}, w_2 \in \Sigma^* \text{ and } |w_1| = |w_2|\}$$

Draw PDA of:

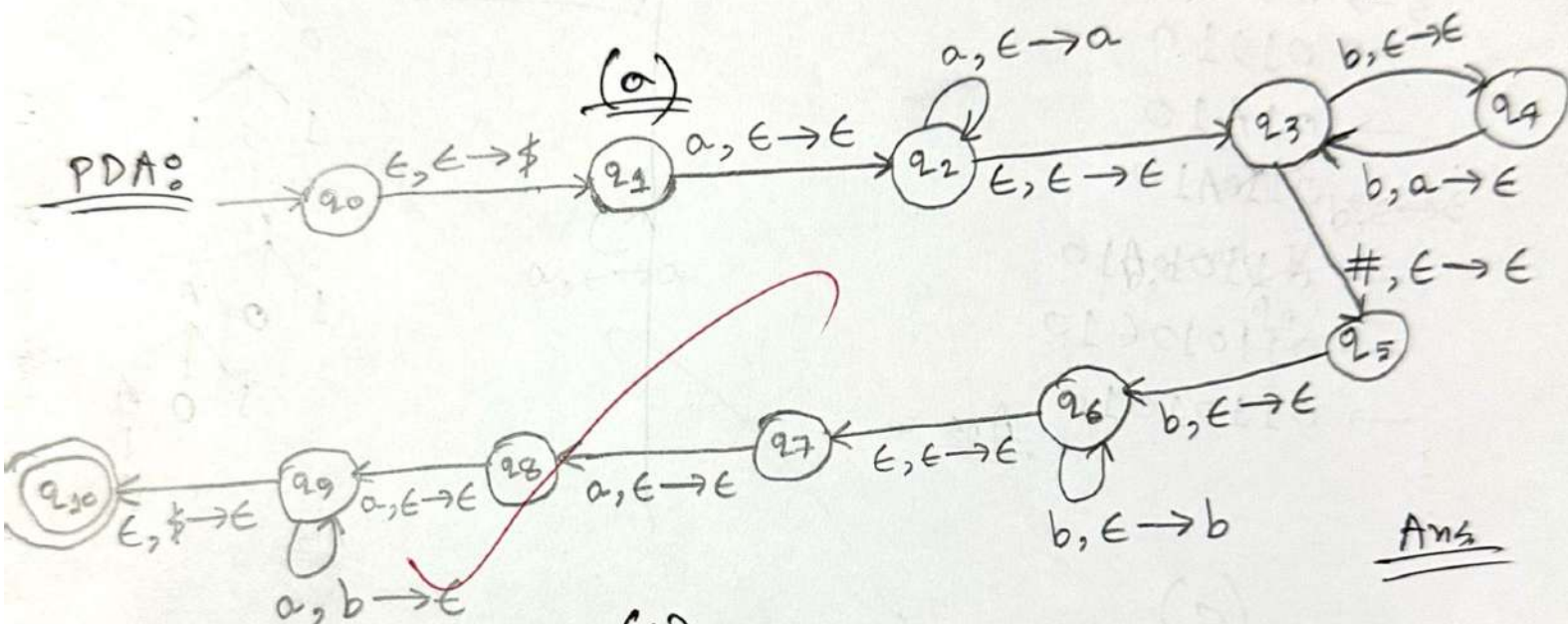
a) L1 ,

b) L2 ,

c) L3

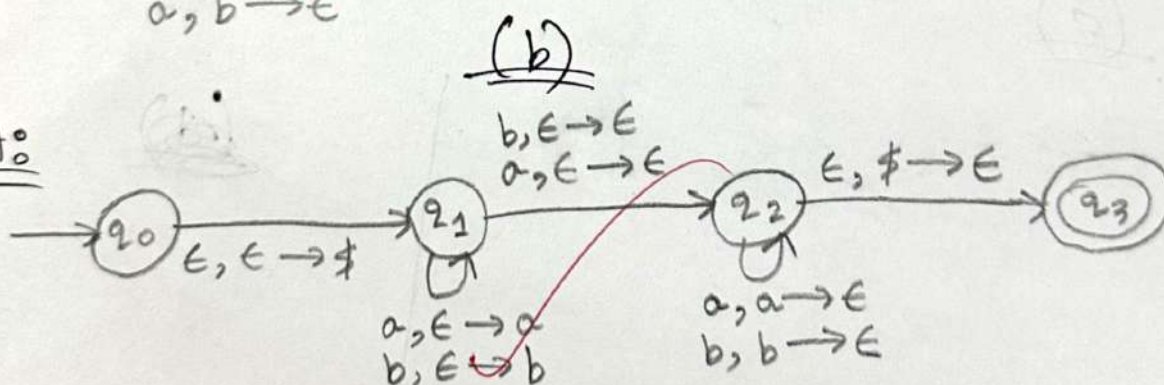
[points: 3+3+4]

PDA:



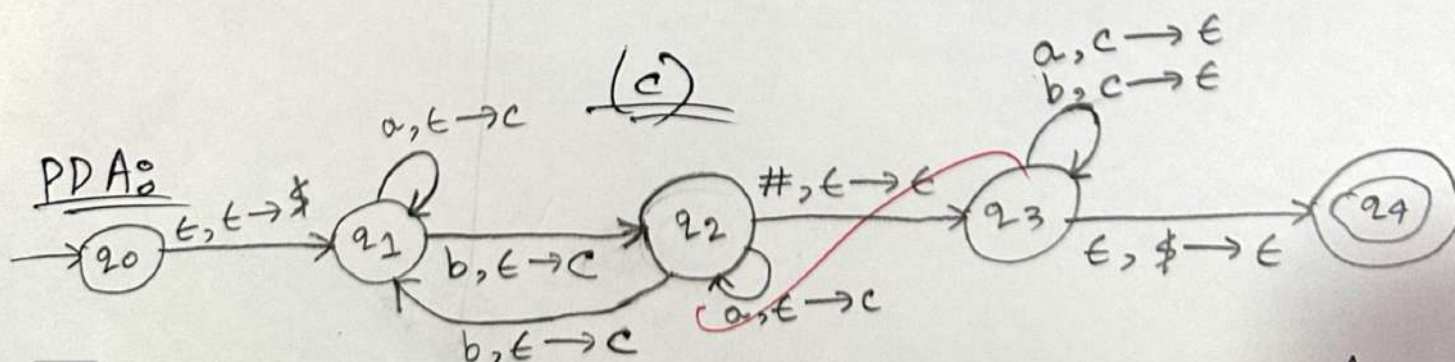
Ans

PDA:



Ans

PDA:



Ans

Question 2: Let $\Sigma = \{0, 1\}$. Consider the following grammar over Σ .

$$S \rightarrow 0S0 \mid 1S1 \mid A$$

$$A \rightarrow 00A \mid 01A \mid 10A \mid 11A \mid \epsilon$$

- (a) Give a leftmost derivation for the string 01101010. (3 points)
 (b) Draw the parse tree corresponding to the derivation you gave in (a). (3 points)
 (c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the given string in (a). (4 points)

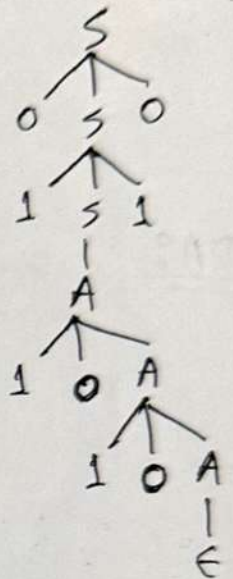
LMD:

$S \rightarrow 0S0$
 $\rightarrow 01S10$
 $\rightarrow 01A10$
 $\rightarrow 0110A10$
 $\rightarrow 011010A10$
 $\rightarrow 011010\epsilon10$
 $\rightarrow 01101010$

Ans.

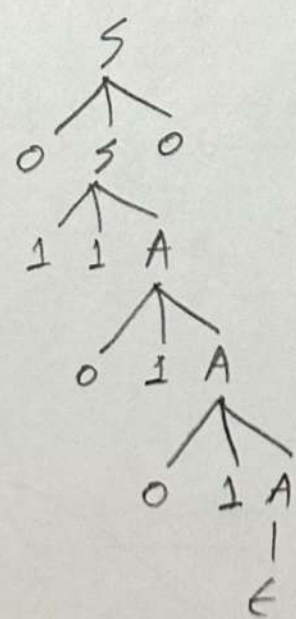
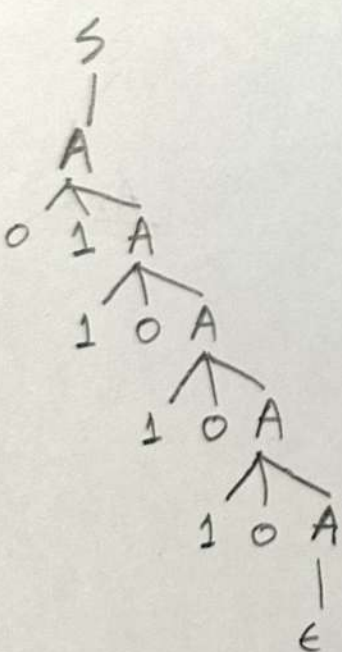
(a)

Parse Tree:



Ans

(c)



Parse Tree 2

Parse Tree 1

$\therefore$ Given Grammar is Ambiguous